

Yonghyeon Jeon

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Research Interests

Numerical Analysis (NA)

- Fractional differential equations, Partial differential equations

Machine Learning (ML)

- Time Series, Machine Learning, Physical-informed Neural Network, Neural Network Architecture

Fractional Calculus (FC)

- Time Series, Caputo, Fractional differential equation

Career

Postdoctoral Research Associate

Mechatronics Research Center, Hongik University

Sejong, South Korea

Sept. 2022 – Aug. 2024

Postdoctoral Research Associate

Department of Mathematical Sciences, UNIST

Ulsan, South Korea

Sept. 2024 – Present

Education

Kyungpook National University

Bachelor of Science, Department of Mathematics

GPA – 3.44/4.0

Adviser: Prof. Eun Sup Kim

Daegu, South Korea

Mar. 2008 – Feb. 2014

Kyungpook National University

Master of Science, Department of Mathematics

GPA – 3.74/4.0

Thesis : A Variable Order Method for Solving Stiff Initial Value Problems

Adviser: Prof. Philsu Kim, Prof. Sang Dong Kim ([link](#))

Daegu, South Korea

Mar. 2014 – Feb. 2016

Kyungpook National University

Ph.D. Mathematics, Department of Mathematics

GPA – 3.84/4.0

*Thesis : Two Approaches of semi – Lagrangian Scheme for Solving
Unsteady Coupled Burgers' Equations*

Advisers: Prof. Philsu Kim ([link](#))

Daegu, South Korea

Mar. 2016 – Aug. 2022

Publication

An economical robust algorithm for solving 1D coupled Burgers' equations in a semi-Lagrangian framework

(with Sangbeom Park, Philsu Kim, and Soyeon Bak), citation: 5
Applied Mathematics and Computation. 2022; 428:127185. ([link](#))

A Novel Decomposition as a Fast Finite Difference Method for Second Derivatives in a semi-Lagrangian framework

(with Soyeon Bak, Sangbeom Park), citation: 0
Results in Mathematics. 2023; 78:22. ([link](#))

Algorithms for coupled Burgers' equations by sharing characteristic curves within BSLM in a semi-Lagrangian framework

(with Soyeon Bak), citation: 0
Advances in Continuous and Discrete Models: Theory and Applications. 2023; 39:2023. ([link](#))

An error predict-correction formula of the load vector in the BSLM for solving three-dimensional Burgers' equations

(with Sangbeom Park, Philsu Kim, and Soyeon Bak), citation: 1
Mathematics and Computers in Simulation. 2024; 221:222–243. ([link](#))

Numerical approach for time-fractional Burgers' equation via a combination of Adams–Moulton and linearized technique

(with Sunyoung Bu), citation: 5
Journal of Mathematical Chemistry. 2024; 62:1189–1208. ([link](#))

Improved numerical approach for Bagley-Torvik equation using fractional integral formula and Adams-Moulton method

(with Sunyoung Bu), citation: 5
Journal of Computational and Nonlinear Dynamics. 2024; 19: 051005. ([link](#))

Deferred correction neural network techniques for solving ordinary differential equations

(with Sunyoung Bu and Kyung Ryeol Baek), citation: 4
Engineering Applications of Artificial Intelligence. 2024; 135: 108771. ([link](#))

Higher-order predictor-corrector methods for fractional Benjamin-Bona-Mahony-Burgers' equations

(with Sunyoung Bu), citation: 4
Journal of Applied Mathematics and Computing. 2025; 71: 1-30. ([link](#))

A semi-implicit predictor-corrector methods for time-fractional Benjamin-Bona-Mahony-Burgers equations

(with Sunyoung Bu), citation: 2
Journal of Mathematical Chemistry. 2025; 63: 526-545. ([link](#))

Higher-order predictor-corrector methods with an enhanced predictor for fractional differential equations

(with Sunyoung Bu), citation: 3
International Journal of Computer Mathematics. 2025; 1–24. ([link](#))

Linearized fractional Adams scheme for Fractional Allen–Cahn equations
(with Sunyoung Bu), citation: 0
Journal of Computational and Nonlinear Dynamics. 2025; 20(6): 061001. (link)

Grant, Scholarship and Award

Supported by basic science research program through the National Research Foundation of Korea (NRF) funded by the Korea government (MSIT) (**grant number RS-2023-00237912**), Jun. 2023 – May. 2026

Talks

- Super convergence method for stiff initial value problems**
2014 KSIAM Annual Meeting, Jeju, Korea, Nov. 2014
- An efficient algorithm of time step control for error correction method**
International Conference
on Applied Mathematics and Numerical Analysis, Bangkok, Thailand, Dec. 2014
- A comparison of multi-step methods and multi-stage methods**
International Eurasian Conference on Mathematical sciences
and applications, Budapest, Hungary, Aug. 2017
- Numerical comparison of one-step method and multi-step method**
KOOK-TAPU Joint Seminar on Knots and Related Topics, Osaka, Japan, Jul. 2019
- A strategy for alleviating order reduction in predictor-corrector methods for FDEs**
The Korean Society for Computational
and Applied Mathematics, Cheongju, Korea, Jun. 2023
- Order reduction mitigating predictor-corrector strategy for FDEs**
The 6th Mediterranean International Conference
of Pure & Applied Mathematics and Related Areas, Paris, France, Aug. 2023
- A corrected PECE algorithm for fractional differential equations**
2024 KSIAM Annual Meeting, Busan, Korea, Dec. 2024
- An Efficient Strategy for Convergence near Initial Singularities
in Time-Fractional Subdiffusion Equations**
KSIAM 2025 Spring Conference, Seoul, Korea, May. 2025
- A heat conduction simulation of a moving heat source subdiffusion model
considering initial weak singularity**
KSIAM 2025 Annual Meeting, Gyeongju, Korea, Nov. 2025

Teaching Experience

Teaching assistant Teaching, grading, office hours	2014–2021
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Teaching	
Calculus I. <i>Teaching, grading, office hours</i>	2023
Teaching	
Calculus I. <i>Teaching, grading, office hours</i>	2023

Computer Skills

MATLAB, MATHEMATICA, C/C++, PHYTHON, L^AT_EX